



FACULTY	Agriculture, Engineering and Natural Sciences		
SCHOOL	Engineering and the Built Environment		
DEPARTMENT	Mechanical and Metallurgical Engineering		
SUBJECT	ENGINEERING MECHANICS II		
SUBJECT CODE	I3641NM		
DATE	Wednesday 21st May 2025		
DURATION	1 Hour 15 Minutes	MARKS	100

Continuous Assessment Test 2

Examiner: Prof. I. O. Ohijeagbon

This question paper consists of **2 pages** including this front page.

Instructions

1. Closed book examination
2. Read the questions carefully
3. This question paper consists of **Three** questions
4. Answer **any 2 Questions**
5. Each question carries **50 marks**
6. Neatly drawn sketches should be used where required
7. **Supplementary materials supplied:** None

Question One

(a) The pulley of an industrial electric motor has an angular acceleration of $\alpha = (1.5\theta) \text{ rad/s}^2$, where θ is in radians. The pulley has a radius of 0.2 m and assumed to start from rest. Determine:

- (i) the magnitude of the velocity and
- (ii) the acceleration of a point **P** located on the edge of the pulley after it has rotated 2 revolutions.

[30 Marks]

(b) If the pulley in Q. 1 (a) drives a wheel of 0.5 m diameter. Determine:

- (i) the speed of the wheel **Q** in revolution per minute (rpm) and
- (ii) the acceleration at a point **m** located at the outermost part of the wheel **Q**.

[20 Marks]

Question Two

(a) A planar kinetic analysis of a machine was conducted. And it was required to understand the effect the machine elements has in resisting angular acceleration. Hence, determine the moment of inertia about the axial axis of the cylinder shown in Figure Q. 2 (a). [10 Marks]

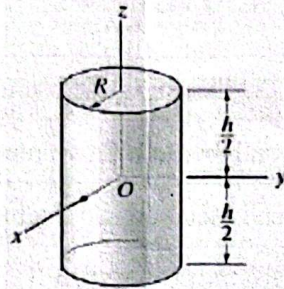


Figure Q. 2 (a)

(b) The pendulum in Figure Q. 2 (b) is suspended from the pin at O and consists of two thin rods, each having a weight of 5 kg . Determine (i) the moment of inertia of the pendulum about an axis passing through point O , (ii) the mass center G of the pendulum relative to point O and (iii) the moment of inertia about G . The moment of inertia of a slender rod shown in Figure Q. 4 (c) is given as: $I_{xx} = I_{yy} = \frac{1}{12}ml^2$; $I_{x'x'} = I_{y'y'} = \frac{1}{3}ml^2$; $I_{z'z'} = 0$ [40 Marks]

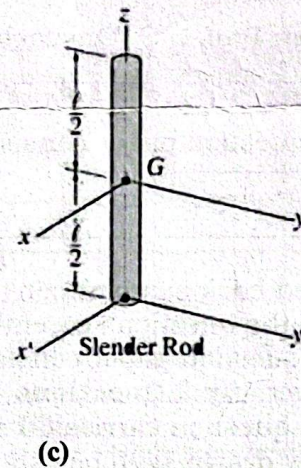
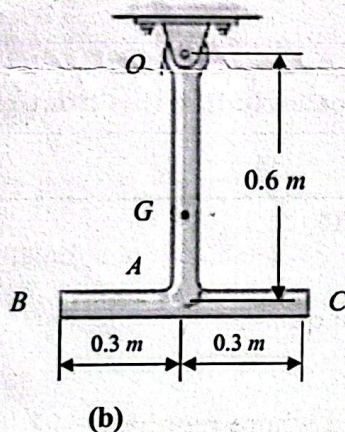


Figure Q. 2 (b)

Question Three

(a) A car A has a mass of $2,000 \text{ kg}$ and a center of mass at G . The distance of the front and rear wheels from G are 1.25 m and 0.75 m respectively while the height of G is 0.3 m . If the mass of the wheels are neglected, and the coefficient of kinetic friction between the wheels and the road is $\mu_k = 0.25$. Determine:

- the acceleration and
- the reactions at the front and rear wheels, if the rear "driving" wheels are always slipping, whereas the front wheels are free to rotate.

[26 Marks]

(b) Assuming a different car B was designed to function conversely to the car A in Question 3 (a). If the front "driving" wheels of car B are always slipping, whereas the rear wheels are free to rotate. Determine:

- the acceleration, and
- the reactions at the front and rear wheels, assuming all other specifications of car B are similar to those of car A .

[24 Marks]